TABLE OF CONTENTS

Page

6. WASTE MANAGEMENT IMPLICATIONS.....1

6.1 Introduction.....1
6.2 Environmental Legislation, Standards and Criteria.....1
6.3 Assessment Methodology.....4
6.4 Identification of Waste Management Implications.....5
6.5 Prediction and Evaluation of Waste Implications.....7
6.6 Mitigation of Adverse Environmental Impacts.....18
6.7 Evaluation of Residual Impacts.....24
6.8 Environmental Monitoring and Audit Requirements.....24
6.9 Conclusion.....25

LIST OF TABLES

Table 6.1	Tentative Construction Programme
Table 6.2	Summary of Estimated Quantities of C&D Materials Generated during Construction Phase
Table 6.3	Summary of Estimated Quantities of C&D Materials Generated during Construction Phase and the Proposed Handling Method / Disposal Outlets
Table 6.4	Classification and Disposal Options for Sediment Samples
Table 6.5	Summary of Estimated Quantities of General Refuse during Construction Phase
Table 6.6	Tentative Transportation Routings for Waste Disposal during Construction Phase
Table 6.7	Summary of Estimated Quantities of Waste to be Generated and Handled during Construction Phase
Table 6.8	Tentative Transportation Routings for Waste Disposal during Operation Phase
Table 6.9	Summary of Estimated Quantities of Waste to be Generated and Handled during Operation Phase

List of Figures

Figure 6.1	Extent of Sediment
Figure 6.2	Locations of Stockpiling Areas

List of Appendices

Appendix 6.1	Extraction of Northern Link EIA Report
Appendix 6.2	Detailed Sediment Categorization

6. WASTE MANAGEMENT IMPLICATIONS

6.1 Introduction

6.1.1.1 This section identifies the types of wastes that are likely to be generated during the construction and operation of the Project and evaluates the potential environmental impacts that may result from the handling, transport and disposal of these wastes.

6.1.1.2 Mitigation measures and good site practices regarding waste handling, storage, collection and disposal have been recommended with reference to the criteria and guidelines as stated in the requirements given in Clause 3.5.6 and Appendix E of the EIA Study Brief (ESB-373/2025), as well as Annexes 7 and 15 of the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM) and relevant waste legislation and guidelines.

6.2 Environmental Legislation, Standards and Criteria

6.2.1 General

6.2.1.1 The criteria and guidelines for assessing waste management implications are outlined in Annexes 7 and 15 of the EIAO-TM, respectively.

6.2.1.2 The following legislation and guidelines also cover the handling, treatment and disposal of wastes in Hong Kong Special Administrative Region (HKSAR):

- Waste Disposal Ordinance (Cap. 354) and subsidiary regulations;
- Waste Disposal (Chemical Waste) (General) Regulation (Cap. 354C);
- Waste Disposal (Charges for Disposal of Construction Waste) Regulation (Cap. 354N);
- Land (Miscellaneous Provisions) Ordinance (Cap. 28);
- Public Health and Municipal Services Ordinance (Cap. 132) – Public Cleansing and Prevention of Nuisances Regulation (Cap. 132 BK); and
- Dumping at Sea Ordinance (Cap. 466).

<u>Waste Disposal Ordinance (Cap. 354)</u>

6.2.1.3 The Waste Disposal Ordinance (WDO) prohibits the unauthorised disposal of wastes. Construction waste is defined under Cap. 354N of the WDO as any substance, matter or thing that is generated and abandoned from construction works regardless of whether or not it has been processed or stockpiled before being abandoned, excluding sludge, screenings or any matter removed or generated from desludging, desilting or dredging works. Under WDO, waste can be disposed of only at designated waste disposal facilities licensed by the Environmental Protection Department (EPD).

<u>Waste Disposal (Chemical Waste) (General) Regulation (Cap. 354C)</u>

6.2.1.4 Issued under the WDO, the Waste Disposal (Chemical Waste) (General) Regulation controls the possession, storage, collection, transport and disposal of chemical wastes. EPD has also issued three guidelines detailing how the Contractor should comply with the regulations on chemical wastes, namely A Guide to the Chemical Waste Control Scheme (updated in 2023), A Guide to the

Registration of Chemical Waste Producers (2024) and Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes (updated in 2023).

Waste Disposal (Charges for Disposal of Construction Waste) Regulation (Cap. 354N)

6.2.1.5 Under the Waste Disposal (Charges for Disposal of Construction Waste) Regulation, construction waste delivered to a landfill for disposal must not contain more than 50% by weight of inert material. Construction waste delivered to a sorting facility for disposal must contain more than 50% by weight of inert material, and construction waste delivered to a Public Fill Reception Facility (PFRF) for disposal must consist entirely of inert material.

Land (Miscellaneous Provisions) Ordinance (Cap. 28)

6.2.1.6 The inert portion of Construction and Demolition (C&D) materials (including rocks, soil, broken concrete, building debris etc.) may be taken to Public Fill Reception Facilities (PFRFs) operated by the Civil Engineering and Development Department (CEDD). The Land (Miscellaneous Provisions) Ordinance requires that individuals or companies who deliver public fill to the PFRFs are required to obtain Dumping Licences. The licences are issued by CEDD under delegated authority from the Director of Lands.

6.2.1.7 Individual licences and windscreen stickers are issued for each vehicle involved. Under the licence conditions, public fill reception facilities will only accept soil, sand, rubble, brick, tile, rock, boulder, concrete, asphalt, masonry or used bentonite. In addition, in accordance with paragraph 12 of the Development Bureau Technical Circular (Works) (DEVB TC(W)) No.6/2010, Public Fill Committee (PFC) will advise on the acceptance criteria. The material will, however, be free from marine mud, household refuse, plastic, metal, industrial and chemical wastes, animal and vegetable matter and any other materials considered unsuitable by the PFRF supervisor.

Public Health and Municipal Services Ordinance – Public Cleansing and Prevention of Nuisances Regulation (Cap. 132BK)

6.2.1.8 The Public Cleansing and Prevention of Nuisances Regulation provides control on illegal dumping of wastes on unauthorised / unlicensed sites. The illegal dumping of wastes can lead to a fine and/or imprisonment.

Dumping at Sea Ordinance (Cap. 466)

6.2.1.9 This Ordinance came into operation in April 1995 and empowers the Director of Environmental Protection (DEP) to control the disposal and dumping of substances and articles at sea for the protection of the marine environment. Under the Ordinance, a dumping permit from the DEP is required for the disposal of regulated substances within and outside the waters of Hong Kong. The permit contains terms and conditions which include the following specifications, but not limited to:

- Type and quantity of substances permitted to be dumped;
- Location of the disposal grounds;
- Requirement of equipment for monitoring the disposal operations; and
- Environmental monitoring requirements.

6.2.1.10 Paragraph 4.2.1 of Chapter 4 of the Project Administration Handbook for Civil Engineering Works (PAH) - Management of Dredged/Excavated Sediment sets out the procedure for seeking approval to and the management framework for marine disposal of dredged / excavated sediment. It outlines the requirements to be followed in assessing and classifying the sediment and explains the marine disposal arrangement for the classified material. Based on paragraph 4.2.1 and Appendix 4.30 of the PAH, there are three types of disposal options for dredged/excavated sediments.

- Type 1 – Open Sea Disposal;
- Type 2 – Confined Marine Disposal; and
- Type 3 – Special Treatment/Disposal.

6.2.2 Other Relevant Environmental Guidelines

6.2.2.1 Other relevant circulars / guidelines applicable to waste management practices for the Project include:

- Works Bureau Technical Circular (WBTC) No. 2/93 - Public Dumps
- WBTC No. 2/93B - Public Filling Facilities
- WBTC No. 16/96 - Wet Soil in Public Dumps
- WBTC No. 12/2000 - Fill Management
- WBTC Nos. 4/98 and 4/98A - Use of Public Fill in Reclamation and Earth Filling Projects
- WBTC No. 19/2001 - Metallic Site Hoardings and Signboards
- WBTC No. 12/2002 - Specifications Facilitating the Use of Recycled Aggregates
- Environment, Transport and Works Bureau Technical Circular (Works) (ETWB TC(W)) No. 19/2005 - Environmental Management on Construction Sites
- Development Bureau Technical Circular (Works) (DEVB TC(W)) No. 06/2010 - Trip Ticket System for Disposal of Construction and Demolition Materials
- DEVB TC(W) No. 08/2010 - Enhanced Specification for Site Cleanliness and Tidiness
- DEVB TC(W) No. 09/2011 - Enhanced Control Measures for Management of Public Fill
- Paragraph 4.1.3 of Chapter 4 of the PAH – Construction and Demolition Materials (subsumed from WBTC Nos. 25/99, 25/99A, 25/99C - Incorporation of Information on Construction and Demolition Material Management in Public Works Subcommittee Papers and ETWB TC(W) No. 33/2002 - Management of Construction & Demolition Material Including Rock);
- A Guide to the Chemical Waste Control Scheme
- A Guide to the Registration of Chemical Waste Producers
- Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes
- Relevant guidelines on handling of yard waste on EPD's website (https://www.epd.gov.hk/epd/english/environmentinhk/waste/manage_facility/ypark.html) and Y·PARK's website (https://www.ypark.hk/zh-hant/);
- The Greening, Landscape and Tree Management Section of the Development Bureau "Guidelines on Yard Waste Reduction and Treatment"; and
- DEVB TC(W) No. 4/2020 "Tree Preservation".

6.3 Assessment Methodology

6.3.1.1 The assessment of waste management impacts arising from the construction and operational activities of the Project has been undertaken in accordance with the EIA Study Brief and criteria given in Annexes 7 and 15 of the EIAO-TM, including the following tasks:

- Identification of the construction and operation activities of the Project which could give rise to waste;
- Estimation of types and quantities of waste generated;
- Examination of opportunities for waste reduction and re-use (both on-site and off-site) and the required disposal options for each type of waste; and
- Evaluation of potential impacts caused by improper handling, collection, transportation and re-use / disposal of wastes with respect to potential hazards, air and odour emissions, noise, wastewater discharges and public transport.

6.3.1.2 Prior to considering the disposal options for various types of waste, opportunities for reducing waste generation, on-site or off-site reuse and recycling have been evaluated. Measures which can be taken in the planning and design phases (e.g. by modifying the design approach), and in the construction phases for maximising waste reduction have been separately considered. In particular, precast prestressed concrete construction method would be employed for the construction of viaduct in general. The precast concrete girder would be manufactured in casting yard offsite and transported to site for final positioning and stitching. Such construction method would minimize the quantity of on-site C&D materials, and extent and duration of construction activities. Precast concrete elements for the construction of Administration, Portal Plant Building / Plant Rooms, and Satellite Control Buildings would be explored wherever possible to minimise on-site work and the associated environmental impacts. Besides, tunnel sections (horseshoe shape) created by the drill-and-blast method minimizes the generation of C&D materials than tunnel sections (circular shape) constructed by tunnel boring machine. Furthermore, slope cutting has been minimized by adjusting the elevation of the road alignment, adoption of a soft ground tunnel, and adopting retaining walls and soil nails such that the extent of slope cutting and the associated generation of C&D materials could be minimized. Practices to promote segregation of waste materials are additionally considered for advancing the waste management efficiency.

6.3.1.3 After considering all the opportunities for reducing waste generation and maximising reuse, the types and quantities of the waste required to be disposed of have been estimated and the disposal options for each type of waste have been described. The disposal method recommended for each type of waste has taken into account the results of the assessment. The impacts caused by handling (including stockpiling, labelling, packaging and storage), collection, transportation and reuse / disposal of waste have been addressed and appropriate mitigation measures have been proposed.

6.4 Identification of Waste Management Implications

6.4.1 Construction Phase

6.4.1.1 The construction activities to be carried out for the Project mainly include site clearance, site formation, tunnelling, piling, excavation works, and demolition work. The potential waste types generated from these construction activities include:

- C&D materials;
- Excavated sediment;
- Chemical waste; and
- General refuse.

6.4.1.2 The construction works of the Project are generally divided into three main phases based on the grouping of major construction activities. The anticipated timing of wastes arising / generated from the major construction activities are shown in **Table 6.1**. In practice, due to the complexity of the Project, certain works across different phases may overlap or be carried out consecutively across phases. The actual construction sequence and schedule will be further optimised based on detailed engineering design. Please refer to the notes below the table for further illustration on the waste generation.

Table 6.1 Tentative Construction Programme

Phase	Tentative Construction Programme	Key Activity Description
Phase 1	Q3 2027 to Q4 2030	Construction advance works and initial civil engineering, including site clearance, at-grade road site formation (comprising temporary haul roads and general site formation), and portal site formation. Later in this phase, main construction works will include like at-grade road construction (including road surfacing), viaduct construction (involving pier foundation excavation), and tunnel excavation etc.
Phase 2	Q1 2031 to Q4 2033	Full scale execution of main civil works which covers the construction of at-grade roads, viaduct construction, tunnel excavation, construction of ancillary structures, etc (e.g., administration buildings, portal plant buildings/ plant room etc.)
Phase 3	Q1 2034 to Q4 2036	Completion of all remaining construction works, including but not limited to finishing of at-grade roads, viaducts, tunnels, and ancillary buildings. It also encompasses the installation, commissioning and testing of associated electrical and mechanical (E&M) systems, as well as site reinstatement works, etc.

Note:

[1] Site clearance in Phase 1 includes the removal of trees and buildings or structures to prepare the area for construction. The activity is likely to carry over to phase 2 and has been reflected in this Chapter.
[2] The viaduct construction activities include piling, pier foundation excavation.
[3] The excavation for wetland compensation has been considered.

6.4.2 Operation Phase

6.4.2.1 The potential waste types generated during the operation of the Project include chemical waste from maintenance activities, and general refuse from the employees in administration buildings.

6.5 Prediction and Evaluation of Waste Implications

6.5.1 Construction Phase

Construction and Demolition (C&D) Materials

6.5.1.1 C&D materials would be generated mainly from site clearance (removal of trees and buildings / structures), site formation (for portal, at-grade road and building), tunnel excavation (e.g. drill-and-blast), pier foundation excavation, road surfacing, underground utilities and wetland compensation excavation works. The C&D materials would comprise both inert C&D materials (e.g. soil, rock and concrete) and non-inert C&D materials (e.g. vegetation, timber, packaging wastes etc.). The non-inert C&D materials would be reused and recycled before disposal to landfill. Yard waste would be sent to the Yard Waste Recycling Centre in Y·PARK for recycling prior to disposal at the designated landfill site. Given that a certain amount of yard waste arising from the construction activities of the Project is anticipated, the waste should be handled in accordance with the principles of reduce, reuse, and recycle (3Rs). During the design of the general layout, devising construction method described in Section 2 and planning of construction programme, considerations have been given to minimise the generation of inert C&D materials and maximise the use of generated inert C&D materials for other construction works. For example, for tunnel construction, drill-and-blast method will be adopted rather than Tunnel Boring Machine (TBM) method in order to minimise the generation of C&D materials from associated tunnel construction works as the tunnel sections created by drill-and-blast would be in the form of a horseshoe surface, whilst that constructed by TBM would be in the form of a circular surface.

6.5.1.2 Based on the latest design information, it is estimated that the total quantity of C&D materials which will be generated during construction phase of the Project would be approximately 5,669,590 m^3 (5,495,370 m^3 of inert materials and 174,220 m^3 of non-inert materials). The major construction activities that generate C&D materials would be carried out from Year 2027 to Year 2036. It should be noted that these quantities are initial estimates only and would need to be further reviewed and updated as the design progresses. Summary of the estimated generation of C&D materials (inert and non-inert) during the construction phase is presented in **Table 6.2**.

Table 6.2 Summary of Estimated Quantities of C&D Materials Generated during Construction Phase

Construction Activities	Estimated Quantities of Non-Inert C&D Materials (m^3) [1]	Estimated Quantities of Inert C&D Materials (m^3) [1]			
		Soft Material	Broken Concrete	Rock	Total
Site Clearance	174,220	0	94,160	0	94,160
Portal Site Formation	0	1,479,590	0	0	1,479,590
At-grade Road Site Formation	0	841,360	0	0	841,360
Tunnel Excavation	0	0	0	2,132,470	2,132,470
Viaduct Pier Foundation Excavation	0	164,020	0	0	164,020
Road Surfacing	0	97,430	0	0	97,430
Building Site Formation	0	184,930	0	0	184,930

Construction Activities	Estimated Quantities of Non-Inert C&D Materials (m^3) [1]	Estimated Quantities of Inert C&D Materials (m^3) [1]			
		Soft Material	Broken Concrete	Rock	Total
Underground Utilities Excavation	0	143,910	0	0	143,910
Wetland Compensation Excavation	0	357,500	0	0	357,500
Total	**174,220**	**3,268,740**	**94,160**	**2,132,470**	**5,495,370**

Note:
[1] The estimated quantities are in-situ volume and rounded to the nearest 10 m^3. Total may not add up due to rounding.

6.5.1.3 Subject to the review at detailed design stage, the inert C&D materials should be reused on-site before being delivered to PFRF as designated by the PFC (such as Tuen Mun Area 38 Fill Bank) for beneficial use in other projects with potential fill demand. Referring to **Table 6.2**, the estimated amount of inert C&D materials generated during the construction phase would be around 5,495,370 m^3, of which around 5,385,470 m^3 is expected to be delivered to PFRF and/or reused in other concurrent projects and around 109,910 m^3 is expected to be reused on-site as backfilling materials. At the early stage of site establishment, the on-site reuse of inert C&D materials will be mainly proposed for site formation and haul road construction, serving the purpose of ground levelling. During the main construction stage, partial backfilling with soil or rock will be required for levelling when constructing U-troughs, retaining walls and abutments.

6.5.1.4 The non-inert C&D materials, upon exhaustion of reuse / recycle effort, would be disposed of at the North East New Territories (NENT) landfill. As shown in **Table 6.2**, the amount of non-inert C&D materials generated during the construction phase is expected to be around 174,220 m^3.

6.5.1.5 A summary of the estimated quantities of C&D materials generated during construction phase and the proposed handling methods / disposal outlets are presented in **Table 6.3**.

Table 6.3 Summary of Estimated Quantities of C&D Materials Generated during Construction Phase and the Proposed Handling Method / Disposal Outlets

Proposed Handling Method / Disposal Outlets of C&D Materials	Estimated Quantities of C&D Materials to be Generated (m^3) [1]		Total C&D Materials (m^3) [1]
	Inert C&D Materials	Non-inert C&D Materials	
Reuse On-site	109,910	-	109,910
Deliver to PFRF and/or other Concurrent Projects for Beneficial Use	5,385,470	-	5,385,470
Disposal at Landfills	-	174,220	174,220
Total C&D Materials	**5,495,370**	**174,220**	**5,669,590**

Notes:
[1] The estimated quantities are in-situ volume and rounded to the nearest 10 m^3. Total may not add up due to rounding.

6.5.1.6 A Construction and Demolition Material Management Plan (C&DMMP) would be prepared in accordance with Section 4.1.3 of the PAH and submitted to PFC for approval. With the implementation of proper management for C&D materials and good site practices as recommended in **Section 6.6.1.5, 6.6.1.6 and 6.6.1.11**, no adverse environmental impacts (including potential hazard, air and odour emissions, noise, wastewater discharges and ecology) and public transport impact caused by handling, collection, transportation and reuse / disposal of C&D materials arising from the Project would be anticipated.

Excavated Sediment

6.5.1.7 Sediment is anticipated to be excavated during the construction of piling and pier for the viaduct section near Fung Kat Heung with reference to **Figure 2.1**. The potential sediment is anticipated from the following:

- Pond sediment;
- Marine deposits.

6.5.1.8 The extent of sediment is provided in **Figure 6.1.** The total volume of sediment generated is estimated to be approximately 11,080 m^3.

6.5.1.9 With reference to the Appendix 8.2 of Northern Link EIA Report (EIA Register No. AEIAR-259/2024), seven drillhole was constructed near Fung Kat Heung in the vicinity of this Project. The extract of EIA report are provided in **Appendix 6.1**.

6.5.1.10 Only 4 of 7 drillhole, namely AUT-EDH(S)01, AUT-EDH(S)02, AUT-EDH(S)03 and 1632-SMA-EDH(S)03, encountered sediment. 8 sediment sample was collected from the 4 drillholes.

6.5.1.11 Testing results have been reviewed in accordance with the Appendix 4.28 of PAH, sample classifications and disposal options as summarized in **Table 6.4**. Detailed sediment categorization is provided in **Appendix 6.2**.

Table 6.4 Classification and Disposal Options for Sediment Samples

Sediment Sample[1]	Depth (m)	Sediment type	Category	Disposal Option[1]
1632-SMA-EDH(S)03	2.0-3.0	Pond deposits	L	Type 1 – Open Sea Disposal
1632-SMA-EDH(S)03	3.0-4.0	Pond deposits	L	Type 1 – Open Sea Disposal
1632-SMA-EDH(S)03	4.0-4.5	Pond deposits	L	Type 1 – Open Sea Disposal
AUT-EDH(S)01	3.5-4.5	Pond deposits	H	Type 2 – Confined Marine Disposal
AUT-EDH(S)02	2.5-3.5	Pond deposits	M	Type 2 – Confined Marine Disposal
AUT-EDH(S)02	3.5-4.5	Pond deposits	M	Type 2 – Confined Marine Disposal
AUT-EDH(S)03	2.5-3.45	Pond deposits	L	Type 1 – Open Sea Disposal
AUT-EDH(S)03	3.5-4.45	Pond deposits	L	Type 1 – Open Sea Disposal

Notes:
[1] Disposal Option according to Appendix 4.28 of PAH.

6.5.1.12 With respect to sediment quality; for pond sediments, metal-based algaecides including copper-based algaecides are not commonly used for freshwater fish farming in Hong Kong due to the high cost and toxicity to fish. Calcium oxide, which is cheaper, non-toxic to fish and a more readily available chemical, is more commonly used to control algae and suspended solids within fish ponds. It is therefore considered unlikely that the pond sediment will be contaminated.

6.5.1.13 The area near Fung Kat Heung was rural before existing road development (San Tin Highway and Tsing Long Highway). Existing estuarine deposits and marine deposits underneath existing land are not likely to be contaminated.

6.5.1.14 Sediment excavated during the project is proposed to be re-used onsite as far as practicable after treatment. Excavated sediment would be treated by cement stabilization / solidification (S/S) and fully reused as backfilling materials within the Project site. Treated sediment will be preferentially reused over general inert C&D materials for backfilling purposes in the Project. In case reuse of sediment is not practical, marine disposal of the sediment will be required according to Section 4.2.1 of the PAH. Should marine disposal of sediment be unavoidable at later stage upon exhaustion of reuse options, separate submissions (e.g. Sediment Sampling and Testing Plan (SSTP) and Sediment Quality Report (SQR)) shall be submitted to EPD's Marine Dumping Control Section / Territorial Control Office if application of dumping permit under DASO is required. The rationale for sediment removal/disposal will also need to be submitted to the Marine Fill Committee (MFC)/CEDD for agreement in accordance with the PAH.

6.5.1.15 The transportation routing and frequency for marine disposal to designated disposal outlets shall be agreed with EPD/ MFC CEDD during application of DASO dumping

permit prior to marine disposal. Detail on transportation vessel will be also submitted for approval during permit application.

Chemical Waste

6.5.1.16 The maintenance and servicing of construction plant, equipment and vehicles involve the use of a variety of chemicals and generate chemical wastes. The possible chemical waste that would be generated during the course of construction works includes:

- Oil and grease associated with plant maintenance;
- Spent hydraulic fluid from plant machinery;
- Scrap batteries from vehicle maintenance; and
- Used paint, cleaners, solvents used in maintaining mechanical equipment.

6.5.1.17 It is difficult to quantify the amount of chemical waste that would arise from the construction activities as it would be highly dependent on the Contractor's on-site maintenance activities and the quantity of plant and equipment utilised. It is anticipated that the quantity of chemical waste, such as lubrication oil and solvent produced from plant maintenance, would be in the order of few hundred litres per month. The amount of chemical waste to be generated would be quantified in the Waste Management Plan (WMP) to be prepared by the Contractor. With the implementation of measures recommended in **Section 6.6.1.19**, no adverse environmental impacts and public transport impact associated with handling, collection, transportation, and disposal of chemical waste is expected.

6.5.1.18 Materials classified as chemical waste shall require special handling and storage arrangements by the Contractor. All chemical waste shall be collected by a licensed collector and be disposed of at a licensed chemical waste treatment and disposal facility such as Chemical Waste Treatment Centre (CWTC) at Tsing Yi via the tentative transportation routing as presented in **Table 6.6**. Unused chemical or those with remaining functional capacity would be reused and recycled on-site or by licensed companies. Mitigation and control requirements for chemical wastes are detailed in **Section 6.6**. Provided that the handling, storage and disposal of chemical wastes are in accordance with these requirements and that stipulated in the Waste Disposal (Chemical Waste) (General) Regulations and the Code of Practice on Packaging, Labelling and Storage of Chemical Wastes published by EPD, adverse environmental impacts (including potential hazard, air and odour emissions, noise, wastewater discharge and ecology) and public transport impact are not anticipated.

General Refuse

6.5.1.19 General refuse comprises of food waste, wastepaper, empty containers etc. would be generated from the workforce during the construction phase of the Project. Improper collection and removal of general refuse would give rise to hygiene problems and adverse environmental impacts, e.g. odour impacts, water pollution, visual impacts, vermin and pest infestation.

6.5.1.20 It is anticipated that the number of construction workers to be working on-site at any one time is estimated to be about 6,300 to 15,300. Based on a generation rate of 0.65 kg per worker per day, the daily arising of general refuse would be about 4,095 kg per day to 9,945 kg per day. The estimated amount of general refuse generated during construction phase is summarised in **Table 6.5**. It is estimated

that around 17,550 tonnes of general refuse would be generated during the construction phase of the Project.

Table 6.5 Summary of Estimated Quantities of General Refuse during Construction Phase

Period	Approx. No. of Workforce	Daily Waste Generation (kg/day)[1]	Duration (Month)	Estimated Quantities of General Refuse Generated (tonnes) [2]
Q3/2027 – Q4/2030	7,200	4,680	42	4,914.0
Q1/2031 – Q4/2033	15,300	9,945	36	8,950.5
Q1/2034 – Q4/2036	6,300	4,095	36	3,685.5
			Total	**17,550.0**

Notes:
[1] Adopt generation rate of 0.65 kg/person/day.
[2] Assume 25 working days per month.

6.5.1.21 The general refuse will be collected on-site on a daily basis, separate from C&D materials, by a reputable waste collector to be employed by the Contractor. The general refuse would be disposed of at strategic landfill, e.g. NENT Landfill. Prior to disposal off-site, the general refuse will be temporarily put in suitably covered storage areas / bins. The covered storage areas / bins will be regularly cleaned and maintained to avoid attracting vermin and pests. With proper on-site handling and storage as well as regular disposal of the waste, no adverse environmental impact (including potential hazard, air and odour emissions, noise and wastewater discharges) or public transport impact would be anticipated. Recommendations of mitigation measures for managing general refuse are presented in **Section 6.6.1.20 to 6.6.1.22**.

Transportation Arrangement for Waste Disposal during Construction Phase

6.5.1.22 Stockpiling areas would be set up within the works areas to handle waste generated during construction phase before disposal. No barging points or conveyor systems would be established in the Project site. The locations of stockpiling areas are shown in **Figure 6.2**. Storage of waste on-site may induce adverse environmental impacts if not properly managed. Mitigation and control measures for the handling and storage of waste are presented in **Section 6.6.1.9 to 6.6.1.11** With the implementation of the recommended mitigation measures, no adverse environmental impacts are expected due to handling and storage of waste on-site.

6.5.1.23 Land transport would be used to deliver and dispose of the waste generated from the Project site to the designated disposal outlets. For transportation routing and frequency of trucks for waste disposal, the Contractor will be requested to use the suitable route to transport waste generated from the Project to the dedicated treatment facilities/disposal sites for disposal. The tentative transportation route is summarised in **Table 6.6**. The transportation routings may change subject to the actual road traffic conditions. Nevertheless, with the implementation of appropriate mitigation measures (e.g. using water-tight containers and covered trucks), no adverse environmental impacts are expected due to the transportation of waste.

Table 6.6 Tentative Transportation Routings for Waste Disposal during Construction Phase

Treatment Facility / Disposal Site / Transfer Point	Type of Waste	Tentative Transporting Routing	Maximum Frequency of Truck Trip Involved[1]
PFRF (such as Tuen Mun Area 38 Fill Bank)	Inert C&D Materials	Via San Tin Highway, Yuen Long Highway, New Territories Circular Road, Tuen Mun Road, Wong Chu Road, Lung Fu Road and Lung Mun Road to Tuen Mun Area 38 Fill Bank	960 trips per day[2][5][6] (During Peak)
Y·PARK / NENT Landfill	Non-inert C&D Materials	Y·PARK (for yard waste): Via San Tin Highway, Yuen Long Highway, New Territories Circular Road, Tuen Mun Road, Wong Chu Road, Lung Fu Road, Lung Mun Road and Lung Kwu Tan Road NENT Landfill: Via San Tin Highway, Fanling Highway, Heung Yuen Wai Highway and Wo Keng Shan Road	23 trips per day[2][3][5][6] (During Peak)
NENT Landfill	General Refuse	Via San Tin Highway, Fanling Highway, Heung Yuen Wai Highway and Wo Keng Shan Road to NENT Landfill	5 trips per day[4][7] (During Peak)
CWTC	Chemical Waste	Via San Tin Highway, Tsing Long Highway, Tsing Sha Highway and Tsing Yi Road to CWTC	1 trip per month

Notes:
[1] Frequency of truck trips are based on assumed capacity of 6.5 m^3 per truck and 25 days per month.
[2] Bulk factor of 1.4 has been applied.
[3] As only trunks in appropriate health condition could be delivered and accepted by Y·Park, the delivered volume and number of trips to Y·Park cannot be determined at this stage. The total number of trips for transporting all non-inert C&D materials to either NENT Landfill or Y·Park has been provided instead.
[4] The assumption of the bulk density of general refuse is 311.73 kg/m^3.
[5] The peak year is projected to be 2029, with an estimated 1,215,035 m^3 of inert C&D materials and 29,036 m^3 of non-inert C&D materials.
[6] A 10% margin has been applied to truck trip calculations to account for programme flexibility.
[7] The peak period will occur between Q1/2031 and Q4/2033.

Summary of Waste to be Generated and Handled during Construction Phase

6.5.1.24 The estimated amount of waste generated during construction phase of the Project, with their proposed handling method / disposal outlets, are summarised in **Table 6.7** below.

6.5.1.25 Trip ticket system would be implemented with reference to DEVB TC(W) No. 06/2010 Trip Ticket System for Disposal of Construction & Demolition Materials, to monitor disposal of waste and to control fly-tipping at PFRFs or landfills. Additionally, all dump trucks and vessels should be equipped with Global Positioning System (GPS) or equivalent system for real time tracking and monitoring of their travel routings and parking locations to prohibit illegal dumping and landfilling of C&D materials or sediments. A recording system for the amount of waste generated, recycled and disposed (including disposal sites) as well as travel routings and parking locations should be proposed.

Table 6.7 Summary of Estimated Quantities of Waste to be Generated and Handled during Construction Phase

Type of Waste	Source	Estimated Quantities				Proposed Handling Method / Disposal Outlets
		Total	On-site Reused / Recycled	Off-site Reused / Recycled	Disposal	
Inert C&D Materials	Materials generated from site clearance, site formation (for portal, at-grade road and building), tunnel excavation, pier foundation excavation, road surfacing, underground utilities and wetland compensation excavation works	5,495,370 m^3	109,910 m^3	5,385,470 m^3	-	• Reuse within the site / deliver to PFRF (such as Tuen Mun Area 38 Fill Bank), subject to allocation by PFC. • Direct matching for reuse in other concurrent projects
Non-inert C&D Materials		174,220 m^3	-	-	174,220 m^3	• Reuse or recycle before disposal at NENT Landfill.
Sediment	Construction of piling and pier for the viaduct section Fung Kat Heung	11,080 m^3	11,080 m^3	-	-	• Treatment by cement S/S technique to treatment standards (UCS ≥ 1000kPa & TCLP ≤ UTS) and reuse on site
Chemical Waste	Oil and grease associated with plant maintenance; Spent hydraulic fluid from plant machinery; Scrap batteries from vehicle maintenance; and Used paint, cleaners, solvents used in maintaining	Few hundred litres per month	-	-	Few hundred litres per month	• Store in compatible containers in designated area on site. • Collect by a licensed collector. • Recycle by licensed facility and/or dispose of at the CWTC.

Type of Waste	Source	Estimated Quantities				Proposed Handling Method / Disposal Outlets
		Total	On-site Reused / Recycled	Off-site Reused / Recycled	Disposal	
	mechanical equipment					
General Refuse	Refuse generated from construction workforce	17,550 tonnes	-	-	17,550 tonnes	• Provide on-site collection points together with recycling bins. • Collect by a reputable waste collector to be employed by the Contractor. • Reuse or recycle before disposal of at NENT Landfill.

6.5.2 Operation Phase

Chemical Waste

6.5.2.1 Chemical wastes such as used paints, lubricants and used batteries may be generated during maintenance and servicing activities (e.g. E&M plant and equipment). This waste may pose environmental, health and safety hazards. Measures as stipulated in the Waste Disposal (Chemical Waste) (General) Regulation and the Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes would be strictly followed for the handling and disposal of chemical waste.

6.5.2.2 Should any chemical waste be generated during operation phase, the operator should register with EPD as a chemical waste producer. The chemical waste should be collected periodically in drum-type containers by licensed chemical waste collectors and disposed of at the CWTC at Tsing Yi.

6.5.2.3 The estimated maximum quantities of chemical waste generated from the operation phase would be about few hundred litres per month. Mitigation and control requirements for chemical waste are detailed in **Section 6.6.2.1**. Provided that the handling, storage and disposal of chemical waste are in accordance with these requirements, no adverse environmental impact associated with storage, handling and disposal of chemical waste during the operation phase of the Project is expected.

General Refuse

6.5.2.4 General refuse will arise from the employees in the administration buildings. General refuse would include food waste, waste paper, plastic bottles, aluminium cans, etc. The storage and handling of these wastes may give rise to environmental impacts. Based on currently available information, the total number of personnel is estimated to be not more than 400. Based on *Monitoring of Solid Waste in Hong Kong – Waste Statistics for 2024* published by EPD, the disposal rate and recovery rate of commercial and industrial (C&I) waste in Year 2024 was 0.53 kg per person per day and 48% respectively. By calculation, the C&I waste generation rate in 2024 was 1.02 kg per person per day. Hence, it is estimated that the maximum amount of general refuse to be generated during the operation phase will be about 408 kg per day.

6.5.2.5 Plastics, papers and other recyclable wastes should be separated from the general refuse and recycled as far as possible. The remaining refuse would be collected by reputable waste collectors for disposal of at NENT Landfill. With the implementation of mitigation measures outlined in **Section 6.6.2.2**, no adverse environmental impact arising from the handling, storage and disposal of general refuse in operation phase would be anticipated.

Transportation Arrangement for Waste Disposal during Operation Phase

6.5.2.6 **Table 6.8** summarised the tentative transportation routing for waste disposal during operation phase.

Table 6.8 Tentative Transportation Routings for Waste Disposal during Operation Phase

Treatment Facility / Disposal Site	Type of Waste	Tentative Transporting Routing	Maximum Frequency of Truck Trip Involved[1]
CWTC	Chemical Waste	Via San Tin Highway, Tsing Long Highway, Tsing Sha Highway and Tsing Yi Road to the CWTC	1 trip per month
NENT Landfill	General Refuse	Via San Tin Highway, Fanling Highway, Heung Yuen Wai Highway, Wong Keng Shan Road to NENT Landfill	1 trip per day

Note:
[1] Assumed truck capacity is 6.5 m^3 per truck and the bulk density of general refuse is 311.73 kg/m^3.

Summary of Waste to be Generated and Handled during Operation Phase

6.5.2.7 **Table 6.9** summarised the estimated amount of waste to be generated and handled during operation phase.

Table 6.9 Summary of Estimated Quantities of Waste to be Generated and Handled during Operation Phase

Type of Waste	Source	Wastes to be Generated	Estimated Quantities to be Generated	Handing Procedures	Handling / Disposal Routes
Chemical Waste	Maintenance and servicing activities, e.g. E&M plant and equipment	Used paints, lubricants and used batteries, etc.	Few hundred litres per month	• Store in compatible containers in designated area on-site • Collect by licensed collectors	Recycle by licensed facility or dispose of at the CWTC
General Refuse	Staff and operation activities within the administration buildings	Food waste, waste paper, plastic bottles and aluminium cans etc.	Maximum 408 kg per day	• Provide on-site collection points together with recycling bins. • Collect by a reputable waste collector.	Reuse or recycle before disposal of at NENT Landfill

6.6 Mitigation of Adverse Environmental Impacts

6.6.1 Construction Phase

Waste Management Hierarchy

6.6.1.1 The waste management hierarchy has been applied in the assessment and development of mitigation measures for waste which aims at evaluating the desirability of waste management methods include the followings in descending order of preference:

- Avoidance and reduction of waste generation;
- Reuse of materials;
- Recovery and recycling of residual materials; and
- Treatment and disposal according to relevant laws, guidelines and good practices.

6.6.1.2 Good site practices and waste reduction measures to achieve avoidance and minimisation of waste generation in the hierarchy are recommended in the following sections. Environmental Management Plan (EMP) and trip ticket system are recommended for monitoring management of waste.

6.6.1.3 Besides, the Contractor is encouraged to deliver waste materials to EcoPark and Y·PARK for reuse and recycling. Referring to latest information in EcoPark's website, EcoPark currently has 12 tenants engaging in recycling business of various waste including waste metals, waste wood, waste plastics, waste lead acid batteries, waste construction material, waste glass, waste rubber tyres and waste paper. The Contractor can consider delivering these waste materials generated during the construction phase to EcoPark for recycling. In general, Y·PARK mainly receives 6m long or below tree trunks and its attached tree branches, twigs and leaves. The Contractor can consider transferring yard waste generated during site clearance to Y·PARK for reuse or recycle after proper treatment. Specifically, in order to minimise the generation of yard waste, the Project Proponent shall:

- Avoid unnecessary removal or excessive pruning of trees. Preserve trees in their original locations and implement tree transplanting when on-site preservation is not feasible;
- Segregate various types of yard waste and shred wood to facilitate reuse and recycling;
- Reuse yard waste on-site for a variety of purposes (e.g., decomposition and composting, recreational and decorative uses, and mulching in planting areas, etc.); and
- Identify recycling options (e.g. delivery to Y·PARK) for yard waste that cannot be directly reused on-site.

6.6.1.4 Where yard waste generation is unavoidable, sorting of yard waste for recycling and reuse on-site should always be the priority. Yard waste shall be separated from C&D material to facilitate recycling, such as delivery them to Y·PARK so as to minimise the quantity of waste to be disposed at landfill site. The Contractor should be responsible for cutting and shredding the yard waste in order to meet the collection requirement of the recycling outlet for processing. Disposal of yard waste directly at landfills should only be regarded as the last resort, when no alternatives are available.

Good Site Practices

6.6.1.5 It is anticipated that adverse waste management implications would not arise in the construction site, provided that good site practices are strictly followed. Recommendations for good site practices during the construction phase include:

- Nomination of approved personnel, such as a site manager, to be responsible for implementation of good site practices, arrangements for waste collection and effective disposal to appropriate facilities;

- Training of site personnel in site cleanliness, concepts of waste reduction, reuse and recycling, proper waste management and chemical waste handling procedures;
- Provision of sufficient waste reception / disposal points of a suitable vermin-proof design that minimises windblown litter;
- Arrangement of regular collection of waste for transporting off-site and final disposal;
- Adoption of appropriate measures to minimise windblown litter and dust during transportation of waste by either covering trucks or by transporting wastes in enclosed containers;
- Provision of regular cleaning and maintenance programme for drainage systems, sumps and oil interceptors;
- Adoption of a recording system for the amount of wastes generated, recycled and disposed (including the disposal sites); and
- Preparation of WMP, as a part of the EMP in accordance with ETWB TC(W) No. 19/2005 - Environmental Management on Construction Sites for submission to the Project Manager for approval.

6.6.1.6 In order to monitor the delivery of C&D materials to the designated PFRF and landfill; and to avoid illegal dumping and control fly-tipping, a trip ticket system should be included with reference to DEVB TC(W) No. 6/2010. A recording system for the amount of waste generated, recycled and disposed, including the disposal sites, should also be set up. CCTV should also be installed at the vehicular entrance and exit of the site as additional measures to prevent illegal dumping / fly-tipping. When delivering inert C&D materials to a PFRF for beneficial reuse, the material shall only consist of soil, rock, concrete, brick, cement plaster / mortar, inert building debris, aggregates and asphalt, and be free from marine mud, household refuse, plastic, metals, industrial and chemical waste, animal and vegetable matter, and other material considered to be unsuitable by the Filling Supervisor. GPS or equivalent systems should also be equipped at all dump trucks and vessels for real-time tracking and monitoring of transportation of inert C&D materials to designated locations as one of the practicable means of avoiding illegal dumpling and landfilling.

Waste Reduction Measures

6.6.1.7 Good management and control of construction site activities / processes can minimise the generation of waste. Waste reduction is best achieved at the planning and design stage, as well as by ensuring the implementation of good site practices. Recommendations to achieve waste reduction are discussed as follow:

- Segregate and store different types of construction related waste in different containers, skips or stockpiles to enhance reuse or recycling of materials and their proper disposal;
- Provide separate labelled bins to segregate recyclable waste such as aluminium cans from other general refuse generated by the work force, and to encourage collection by individual collectors;
- Recycle any unused chemicals or those with remaining functional capacity;
- Maximise the use of reusable steel formwork to reduce the amount of C&D materials;
- Wood, steel and other metals shall be separated for re-use and / or recycling prior to disposal of non-inert C&D materials to minimise the quantity of waste to be disposed of to landfill;

- Adopt proper storage and site practices to minimise the potential for damage to, or contamination of construction materials;
- Plan the delivery and stock of construction materials carefully to minimise the amount of surplus waste generated;
- Adopt pre-cast construction method instead of cast-in-situ method for construction of concrete structures; and
- Minimise over-ordering and wastage through careful planning during procurement of construction materials.

6.6.1.8 In addition to the above good site practices and waste reduction measures, specific mitigation measures are recommended below for the identified waste so as to minimise environmental impacts during handling, transportation and disposal of these wastes.

Storage, Collection and Transportation of Waste

6.6.1.9 Storage of materials on-site may induce adverse environmental impacts if not properly managed. The following recommendations should be implemented to minimise the impacts:

- Waste, such as soil, should be handled and stored well to ensure secure containment, thus minimising the potential of pollution;
- Storage areas should be maintained and cleaned routinely;
- Stockpiling area should be provided with covers and water spraying system to prevent materials from wind-blown or being washed away; and
- Different locations should be designated to stockpile each material type to enhance reuse.

6.6.1.10 Waste collectors should be employed for the collection and transportation of waste generated. For chemical waste, licensed waste collector for its transportation and disposal should be employed in accordance with the Waste Disposal (Chemical Waste) (General) Regulation. The following recommendations should be implemented to minimise the impacts:

- Remove waste in a timely manner;
- Employ trucks with cover or enclosed containers for waste transportation;
- Obtain relevant waste disposal permits from the appropriate authorities;
- Dispose waste at licensed waste disposal facilities; and
- Maintain records of quantities of waste generated, recycled and disposed.

Construction and Demolition Material

6.6.1.11 Subject to site situation, the generated C&D materials should be sorted on-site into inert C&D materials and non-inert C&D materials. It is the Contractor's responsibility to separate the inert and non-inert C&D materials on-site. To minimise the impact resulting from collection and transportation of C&D materials, non-inert C&D materials, such as wood, plastic, steel and other metals should be reused or recycled and, as a last resort, disposed to landfill. Inert C&D materials should be recycled and reused. With reference to **Table 6.7**, surplus inert C&D materials are expected to be delivered to PFRF (e.g. Tuen Mun Area 38 Fill Bank), subject to allocation by PFC and/or reused in other concurrent projects. Within the stockpiling areas, the following measures should be taken to control potential environmental impacts or nuisance:

- Proper handling and storage of waste by means of covers and/or water spraying system to minimise the potential environmental impact and to prevent materials from wind-blown or being washed away;
- Covering materials during heavy rainfall;
- Locating stockpiles suitably so as to minimise potential visual impacts;
- Displaying warning signs at the designated disposal sites. CCTV should be installed at the vehicular entrance and exit of the site as additional measures to prevent fly-tipping;
- Adopting GPS or equivalent system for tracking and monitoring of all dump trucks for the Project to record their travel routings and parking locations to prohibit illegal dumping and landfilling of C&D materials; and
- Minimising land intake of stockpile areas, subject to site condition.

Excavated Sediment

6.6.1.12 Excavated sediment should handled, treated, transported and/or disposed of in a manner that would minimize adverse environmental impacts. Beneficial reuse will be considered before the disposal of excavated sediment. In order to minimise off-site disposal, the excavated sediment is proposed to be reused on site under the Project (e.g. as backfilling materials). The excavated sediment would be treated by cement stabilization / solidification (S/S) to improve the material strength to render it suitable for reuse. Treated materials will be subject to Unconfined Compressive Strength (UCS) and Toxicity Characteristic Leaching Procedure (TCLP) testing with a UCS acceptance criteria of ≥ 1000kPa and TCLP meeting the Universal Treatment Standards (UTS) in Title 40 of the Code of Federal Regulations (CFR). If treated materials do not meet the UCS and TCLP acceptance criteria, the materials will be re-treated and re-tested. Pilot-scale trials should be carried out before the commencement of treatment to determine the cement/sediment ratio and curing time to achieve the necessary strength. Off-site sediment disposal should only be considered as the last resort in case sediment treatment / reuse not possible and should follow the procedures in the PAH.

6.6.1.13 Requirements of the Air Pollution Ordinance (Construction Dust) Regulation, where relevant, shall be adhered to during excavation, treatment, transportation and/or disposal of the sediment.

6.6.1.14 Excavated sediment treatment areas for carrying out the cement S/S process and any temporary stockpiling should be confined. Runoff should be collected and discharged according to the Water Pollution Control Ordinance (WPCO).

6.6.1.15 To minimise the potential odour / dust emissions during excavation and transportation of the sediment, the excavated sediments shall be wetted during excavation / material handling and shall be properly covered when placed on trucks or barges. Loading of excavated sediments to barges (if used) shall be controlled to avoid splashing and overflowing of the sediment slurry to the surrounding water.

6.6.1.16 Should marine disposal of sediment be unavoidable at later stages upon exhaustion of reuse options, separate submissions (e.g. Sediment Sampling and Testing Plan (SSTP) and Sediment Quality Report (SQR)) shall be submitted to EPD's Marine Dumping Control Section / Territorial Control Office if application of a dumping permit under DASO is required. The rationale for sediment removal/disposal will also need to be submitted to the Marine Fill Committee (MFC)/CEDD for agreement in accordance with PAH, **Section 4.2.1**.

6.6.1.17 Stockpiling of sediment requiring Type 2 – Confined Marine Disposal or Type 3 – Special Treatment / Disposal (if any) should be avoided as far as possible. If temporary stockpiling of these sediments is necessary, the excavated sediment should be covered by tarpaulin and the area should be placed within earth bunds or sand bags to prevent leachate from entering the ground, nearby drains and surrounding water bodies. The stockpiling areas should be completely paved or lined in order to avoid contamination to underlying soil or groundwater. Separate and clearly defined areas should be provided for stockpiling of contaminated and uncontaminated materials. Leachate, if any, should be collected and discharged according to the WPCO.

6.6.1.18 Barge transporting sediments to designated disposal sites should be equipped with tight fitting seals to prevent leakage and should not be filled to a level that would cause overflow of materials during loading or transportation. In addition, monitoring of the barge loading shall be conducted to ensure that loss of material does not take place during transportation. To avoid illegal dumping, all dumping vessels have to be approved in a marine dumping permit issued under the DASO. Each of the vessels has to be installed with an automatic recording equipment, namely the Front End Mobile Unit (FEMU), which is a key component of the Real Time Tracking & Monitoring of Vessel (RTTMV) System of EPD. The FEMU transmits self-monitoring data direct from the barge at sea to the Control Centre at EPD through GPRS mobile communication network. The transportation route avoiding the ecological sensitive areas should be proposed when applying the dumping permit.

Chemical Waste

6.6.1.19 For those processes which will generate chemical waste, alternatives should be considered to eliminate/minimise the use of chemicals, to reduce the generation quantities or to select a chemical type of less impact on environment, health and safety. If chemical waste is produced at the construction site, the Contractor would be required to register with the EPD as a Chemical Waste Producer and must follow the guidelines stated in the Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes. Good quality containers compatible with the chemical wastes should be used, and incompatible chemicals should be stored separately. Appropriate labels should be securely attached on each chemical waste container indicating the corresponding chemical characteristics of the chemical waste, such as explosives, flammable, oxidising, irritant, toxic, harmful, corrosive, etc. The Contractor shall employ a licensed collector to transport and dispose of the chemical wastes at a licensed chemical waste treatment and disposal facility such as CWTC at Tsing Yi in accordance with the *Waste Disposal (Chemical Waste) (General) Regulation*.

General Refuse

6.6.1.20 General refuse should be stored in enclosed bins or compaction units separate from C&D materials and chemical wastes. A reputable waste collector should be employed by the Contractor to remove general refuse from the site, on a daily basis to minimise hygiene problems, odour nuisance, or pest infestation. An enclosed and covered area is preferred to reduce the occurrence of wind-blown light materials.

6.6.1.21 The collected general refuse will be disposed of at designated landfill. Clearly labelled recycling bins should be provided on site in order to encourage segregation and recycling of aluminium cans, plastic wastes, and wastepaper in order to reduce general refuse production.

6.6.1.22 The contractor should carry out an education programme for workers in avoiding, reducing, reusing and recycling of materials generation. Posters and leaflets advising the use of the bins should also be provided on site as reminders. The recyclable waste materials should then be collected by reliable waste recycling agents on a regular basis.

6.6.2 Operation Phase

Chemical Waste

6.6.2.1 Generation of chemical waste should be minimised. For those processes which will generate chemical waste, alternatives should be considered to eliminate the use of chemicals, to reduce the generation quantities or to select a chemical type of less impact on environment, health and safety. Similar to construction phase, chemical waste generated should be stored in appropriate containers and collected by a licensed chemical waste collector. Chemical wastes such as spent lubricant oil should be recycled at an appropriate facility. Chemical waste that could not be recycled should be disposed of at licensed chemical waste treatment and disposal facility such as the CWTC at Tsing Yi in accordance with the *Waste Disposal (Chemical Waste) (General) Regulation*.

General Refuse

6.6.2.2 Recyclable materials (e.g. paper, plastic bottles, glass bottles, and aluminium cans) should be segregated on site as far as practicable for off-site recycling. Remaining non-recyclable refuse should be disposed of at NENT Landfill, separate from chemical wastes, on a daily basis by a reputable waste collector to minimise odour nuisance and pest infestation.

6.7 Evaluation of Residual Impacts

6.7.1.1 With the implementation of the recommended mitigation measures for the handling, transportation and disposal of the identified waste arising, residual impacts would not be expected during the construction and operation of the proposed Project.

6.8 Environmental Monitoring and Audit Requirements

6.8.1 Construction Phase

6.8.1.1 It is the Contractor's responsibility to ensure that all wastes produced during the construction of the Project are handled, stored and disposed of in accordance with good waste management practices and EPD's regulations and requirements. The recommended mitigation measures should form the basis of the site WMP to be developed by the Contractors as part of the EMP in accordance with *ETWB TC(W) No. 19/2005 Environmental Management on Construction Sites* and submitted to the Project Manager for approval at the construction stage. The monitoring and auditing requirements stated in *ETWB TC(W) No.19/2005* should be followed with regard to the management of C&D materials.

6.8.1.2 Monthly site audit should be conducted by the Environmental Team (ET) during construction phase to check if wastes are being managed in accordance with approved procedures. The audits should cover site inspection on all aspects of on-site waste management practices including waste generation, storage, recycling, transport and disposal. Apart from site inspections, waste management related

documents including licences, permits, disposal and recycling records should be reviewed and audited for compliance with the legislation of the recommended good site practice and other waste management mitigation measures under the site audit.

6.8.1.3 Besides, auditing on the implementation of the trip ticket system in accordance with *DEVB TC(W) No. 06/2010* and incorporate the requirements as appropriate should also be required.

6.8.2 Operation Phase

6.8.2.1 Adverse environmental impacts would not be anticipated with the implementation of good waste management practices following the existing practices of regular maintenance works. Environmental audit programme on waste management for the operation phase of the Project would not be required.

6.9 Conclusion

6.9.1.1 During construction phase, waste generated from the Project include C&D materials from construction activities, chemical wastes from maintenance and servicing of construction plants and vehicles, general refuse from workforce and excavated sediments. Inert C&D materials are recommended to be reused on-site / other concurrent projects to minimise the amount of inert C&D materials delivered off-site to PFRF. Wastes generated should be handled, transported and disposed of according to the recommended good site practices and measures. With the implementation of these measures, no adverse environmental impacts (including potential hazard, air and odour emissions, noise and wastewater discharges) would be anticipated during the construction phase.

6.9.1.2 The main types of waste generated from the operation of the Project are general refuse and chemical waste. With the implementation of management control practices, no adverse environmental impact would be anticipated.